TITLE
Literature Review

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Declaration

I, Sashin Samlal hereby declare the contents of this research document to be my own work. This document is submitted for degree purposes in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.
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Chapter 1

Introduction

The barrier to entry for the RoboCup Small Size League (SSL) is primarily defined by the prohibitive cost of specialized robotic hardware and the inherent complexity of high-level algorithm implementation. These limitations often exclude researchers and institutions with limited funding from contributing to the advancement of multi-agent systems and competitive robotics.In a South African context, where a significant portion of national resources is prioritized for socioeconomic stability, the high capital requirement of SSL research effectively segregates talented engineers and scientists from global participation. This economic disparity hinders the field's potential growth and restricts the broader societal impact of robotics innovation.A critical solution to this cost barrier lies in the development of a standardized, competition-capable robotic platform that utilizes additive manufacturing and open-source middleware to simplify hardware-software integration. Such a design provides an accessible entry point, fostering wider participation in the SSL and stimulating broader interest in autonomous systems. Historically, low-cost approaches have relied on a "hobbyist" paradigm, utilizing off-the-shelf boards and structural components sourced from general hardware suppliers, such as sheet metal. While various open-source middleware solutions exist to facilitate software standardization, integrating disparate hobbyist components remains a significant challenge due to proprietary communication protocols and manufacturer-specific constraints. Furthermore, traditional manufacturing methods like welding or metalwork require specialized skills to meet the strict dimensional tolerances mandated by league regulations. However, the rapid advancement and cost-effectiveness of 3D printing technology now allow for the production of standardized, modular, and easily reproducible structural components. By integrating this hardware with the Robot Operating System (ROS) middleware, it is possible to create a standardized interface that abstracts low-level control and enables seamless transition between simulation and physical deployment.The establishment of this unified hardware-software framework further permits the implementation of standardized benchmarking suites, providing a rigorous environment for the performance evaluation of diverse robotic components and autonomous strategies.

Chapter 2

Background

2.1 Technical Constraints and Environmental Standards

The RoboCup Small Size League (SSL) represents a highly constrained environment designed to promote research in fast-paced multi-robot systems. To understand the benchmarking of a custom low-cost platform against standard solutions, it is necessary to establish the regulatory and physical parameters of the league.

2.1.1 League Environment and Operational Parameters

The SSL is divided into Division A and Division B, with the latter specifically serving as an entry-point for teams looking to overcome the high mechanical and financial barriers of entry RoboCup Small Size League Technical Committee [2026].

- **Playing Surface:** Matches are conducted on a standardized green carpeted area. Division B utilize a $9m \times 6m$ playing field RoboCup Small Size League Technical Committee [2026].
- **The Ball:** Gameplay utilizes an orange golf ball with a diameter of $0.043m$ and a mass of approximately $0.046kg$. This provides the baseline for all ball manipulation and kicking calculations RoboCup Small Size League Technical Committee [2026].
- **Autonomy:** Human intervention is strictly prohibited during gameplay. All motion planning and strategy must be executed autonomously by off-field computers via wireless communication RoboCup Small Size League Technical Committee [2026].

2.1.2 Mechanical and Hardware Dimensionality

The league enforces strict volumetric constraints to ensure fairness and safety during high-speed maneuvers.

- **Cylindrical Volume:** A robot must at all times fit within a cylinder $0.18m$ in diameter and $0.15m$ in height **?**.

- **Ball Interaction:** Dribbling devices may apply backspin but cannot elevate or fully capture the ball. At least 80% of the ball's area must remain outside the robot's convex hull to prevent "ball hoarding" RoboCup Small Size League Technical Committee [2026].
- **Kicking Velocity:** For safety, the ball velocity is capped at $6.5m/s$ in 3D space, which dictates the design limits of solenoids or pneumatic kicker systems RoboCup Small Size League Technical Committee [2026].

2.1.3 Kinematics and Motion Constraints

Navigation within the SSL requires managing both individual unit kinematics and league-mandated safety pauses.

- **Emergency and Controlled Pauses:** Under a "Halt" command, robots must brake within 2 seconds. During a "Stop" command, speeds are restricted to a maximum of $1.5m/s$ with a minimum distance of $0.5m$ maintained from the ball RoboCup Small Size League Technical Committee [2026].
- **Differential vs. Holonomic Drive:** While competitive teams frequently utilize omnidirectional drives for simultaneous rotation and translation, this study benchmarks a differential-drive system. The differential drive simplifies mechanical complexity by utilizing two independent wheels but introduces non-holonomic constraints that must be accounted for in the ROS-compatible control stack.

2.1.4 Global Vision and Software Architecture

The software architecture relies on a "software glue" approach where centralized vision is distributed to individual agents.

- **SSL-Vision:** A centralized overhead camera system tracks standard dot patterns on top of each robot (within an $0.085m$ radius) to determine unique IDs and coordinates RoboCup Small Size League Technical Committee [2026].

Chapter 3

Related Work

The development of robotic platforms for competitive environments like the RoboCup Small Size League (SSL) has evolved through distinct shifts in locomotion strategy, software architecture, and manufacturing philosophy. This section explores these transitions to motivate the design of a low-cost, 3D-printed benchmarking platform.

3.0.1 Locomotion and Drive Mechanisms

The high-speed, dynamic nature of robotic soccer has traditionally necessitated the use of complex holonomic drives to allow for simultaneous translation and rotation. Platforms designed for the SSL overwhelmingly utilize omnidirectional drive mechanisms to achieve this unconstrained movement. For instance, Smit [2011] developed a four-wheel omnidirectional drive system specifically to replace older, less maneuverable architectures. Similarly, the *First Order* team recently underwent a first-principles mechanical redesign to implement a compact four-wheel omnidirectional base for enhanced agility Liang *et al.* [2025].

However, a significant alternative exists for research conducted in constrained environments. While differential drive systems lack holonomic movement, their kinematic equations allow for zero-radius turning, making them highly capable in tight spaces Chavan and Minase [2015]. This project leverages the differential drive approach, following the success of the *MICROMVP* platform, which utilizes a simple two-wheel system to drastically reduce mechanical complexity and cost while remaining fully capable of validating complex path-planning algorithms Yu *et al.* [2017].

3.0.2 Evolution of System Architecture and Middleware

A critical shift in the field involves the transition from centralized off-field control to distributed onboard intelligence. Traditional SSL systems relied heavily on centralized computers to process vision and command robots. Smit [2011] challenged this by outfitting robots with Single Board Computers (SBCs) running embedded Linux and utilizing the "Player" middleware to create a distributed network.

Modern research has since converged on the Robot Operating System (ROS) as the industry standard. As detailed in Joseph [2018], ROS provides a decentralized, node-based publisher/subscriber architecture that handles complex interprocess communi-

cation via standard library nodes. By adopting this architecture, this project abstracts low-level hardware control, allowing the robot to interface seamlessly with high-level autonomous algorithms.

3.0.3 Accessibility through Cost Optimization

The high barrier to entry in robotics is often defined by manufacturing complexity. To overcome this, recent literature highlights a move away from expensive machined metal toward additive manufacturing. Teams have increasingly utilized 3D-printed plastics to reduce costs and increase design flexibility for complex components Liang *et al.* [2025].

This accessibility is further enhanced by combining 3D printing with off-the-shelf electronics. The *MICROMVP* platform demonstrates that a highly affordable micro-scale testbed is achievable through this combination Yu *et al.* [2017]. Furthermore, Joseph [2018] provides a framework for bridging high-level ROS nodes running on an SBC directly to low-level, inexpensive microcontrollers, an approach central to this project's hardware design.

3.0.4 Bridging the Gap: Simulation vs. Physical Validation

Validation of multi-agent algorithms often occurs in virtual environments to bypass the costs of physical testing. For the RoboCup Middle Size League, researchers developed a simulation system based on ROS and Gazebo, utilizing a custom model plugin to mathematically simulate locomotion Yao *et al.* [2015].

While such simulations simplify robots into basic cylindrical collision elements to preserve computational speed, they require the injection of Gaussian noise to mimic real-world sensor inaccuracies Yao *et al.* [2015]. This "reality gap" motivates the necessity of the current project: by building a physical benchmarking platform, we can quantitatively evaluate how algorithms perform against real-world friction, battery dynamics, and discretization delays that simplified virtual models often fail to capture within the constraints defined by the official league rules RoboCup Small Size League Technical Committee [2026].

Chapter 4

Conclusion

This literature review has explored the evolving landscape of robotic platforms within the RoboCup Small Size League, focusing on the trade-offs between mechanical complexity and research accessibility. The synthesis of current research reveals a significant "reality gap" where high-performance, holonomic competition units remain prohibitively expensive and mechanically complex for emerging teams, while virtual simulations often fail to capture the nuances of physical friction and sensor noise.

The review identified four key pillars essential for modern robotics research: the shift from omnidirectional dominance to viable differential drive alternatives for constrained environments; the transition from centralized off-field processing to distributed, ROS-compatible software architectures; the utilization of additive manufacturing (3D printing) to lower financial barriers; and the critical need for physical benchmarking against industry standards like the TurtleBot v2.

In conclusion, there is a clear need for a standardized, low-cost development platform that bridges the gap between simulation and competition-ready hardware. By integrating off-the-shelf electronics with a modular 3D-printed shell and a robust ROS middleware, the proposed project addresses the identified mechanical roadblocks. This platform will not only provide a credible iteration for quantitative benchmarking but also serves as a replicable entry point for talented researchers in resource-constrained regions, ultimately fostering broader participation and innovation within the field of multi-agent robotics.

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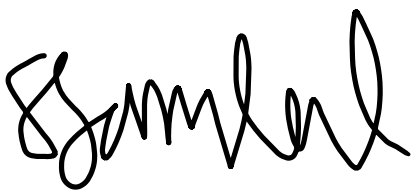
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